Final Individual Project
One-page Summary

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Research question

Is there a significant difference in mean height among students with different levels of milk drinking frequency?

Group Variable

- Group variable: NoMilkDrinking
- Definition: This variable classifies students into three groups for analysis, where 0 = No milk drinking, 1 = Occasional milk drinking, and 2 = Frequent milk drinking.

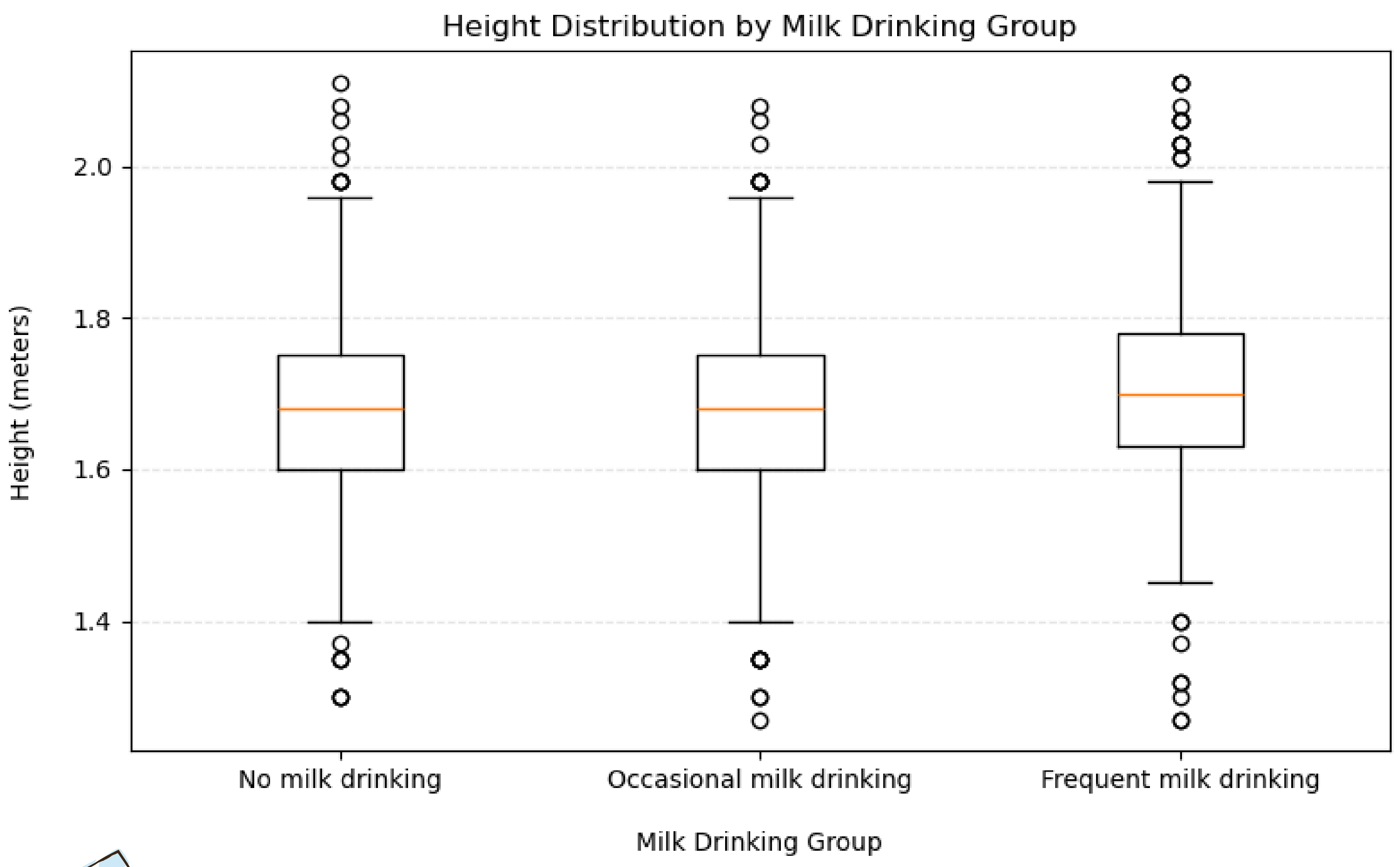
Response variable

- Response variable: HowTallAreYouWithoutShoesInMeters
- Definition: This variable records students' height without shoes in meters.

Method

1.One-Way ANOVA
2.Tukey HSD post-hoc test

Figure

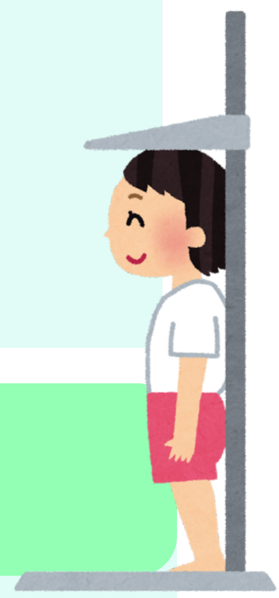


Table

Comparison	Group 1	Group 2	Mean Difference
Comparison 1	0 = No milk drinking	1 = Occasional milk drinking	0.0104
Comparison 2	0 = No milk drinking	2 = Frequent milk drinking	0.0364
Comparison 3	1 = Occasional milk drinking	2 = Frequent milk drinking	0.0259

Key result

- The null hypothesis (H_0: $\mu_1 = \mu_2 = \mu_3$) was rejected at $\alpha = 0.05$.
- The one-way ANOVA indicated that at least one group mean height differed among the three groups ($F(2, 12665) = 139.056$, $p < 0.001$).
- Tukey HSD post hoc tests showed significant pairwise differences among all groups.
- Descriptive statistics indicated that frequent milk drinkers had the highest mean height, followed by occasional milk drinkers and non-drinkers.



Conclusion

There is sufficient statistical evidence at the 0.05 significance level to conclude that at least one group mean height differs among the three milk drinking groups.